



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Trigonometric Graphs & Equations

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- * Solving Trig Equations



Your notes

Trigonometric Graphs

Drawing Trig Graphs

What are trig graphs?

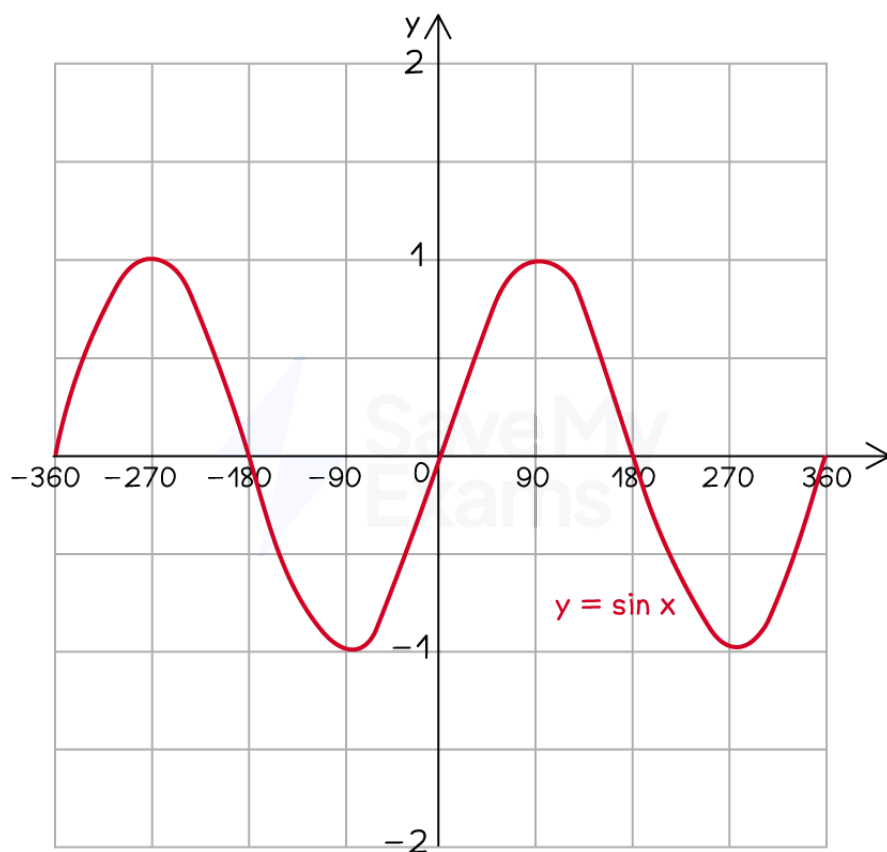
- Trigonometric (trig) graphs are the graphs of
 - $y = \sin x$
 - $y = \cos x$
 - $y = \tan x$
- The variable x is like an angle
 - but the angle can now go **beyond acute** to become **obtuse** and **reflex**
 - $0^\circ \leq x \leq 360^\circ$
- Trig graphs have **repeating** (periodic) shapes and **symmetries** that you need to know

How do I draw the graph of $y = \sin x$?

- The graph of $y = \sin x$ is a **wave** that oscillates between **heights of 1 and -1** and **repeats every 360°** (its **period** is 360°)
 - It goes through the **origin**, $(0, 0)$
 - Then every 90° it cycles through the heights 1, 0, -1, 0, ...



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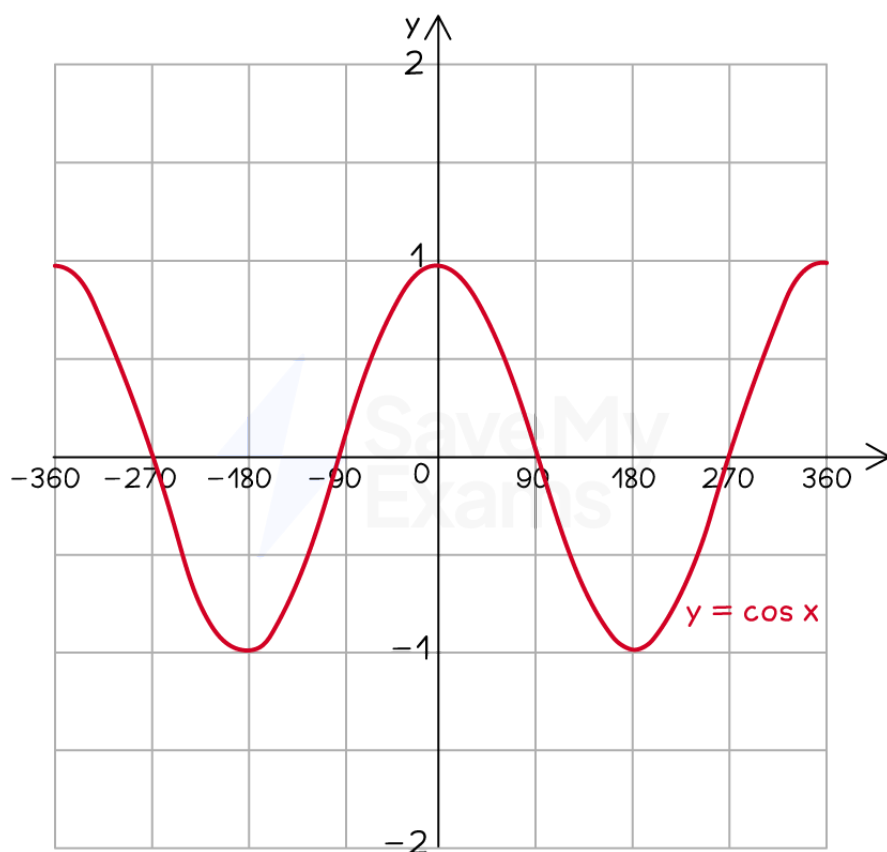
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How do I draw the graph of $y = \cos x$?

- The graph of $y = \cos x$ is a **wave** that oscillates between **heights of 1 and -1** and **repeats every 360°** (its **period** is 360°)
 - It has a **y-intercept of 1**, coordinates (0, 1)
 - Then every 90° it cycles through the heights 0, -1, 0, 1, ...
- $y = \cos x$ is the same as translating $y = \sin x$ by 90° to the left



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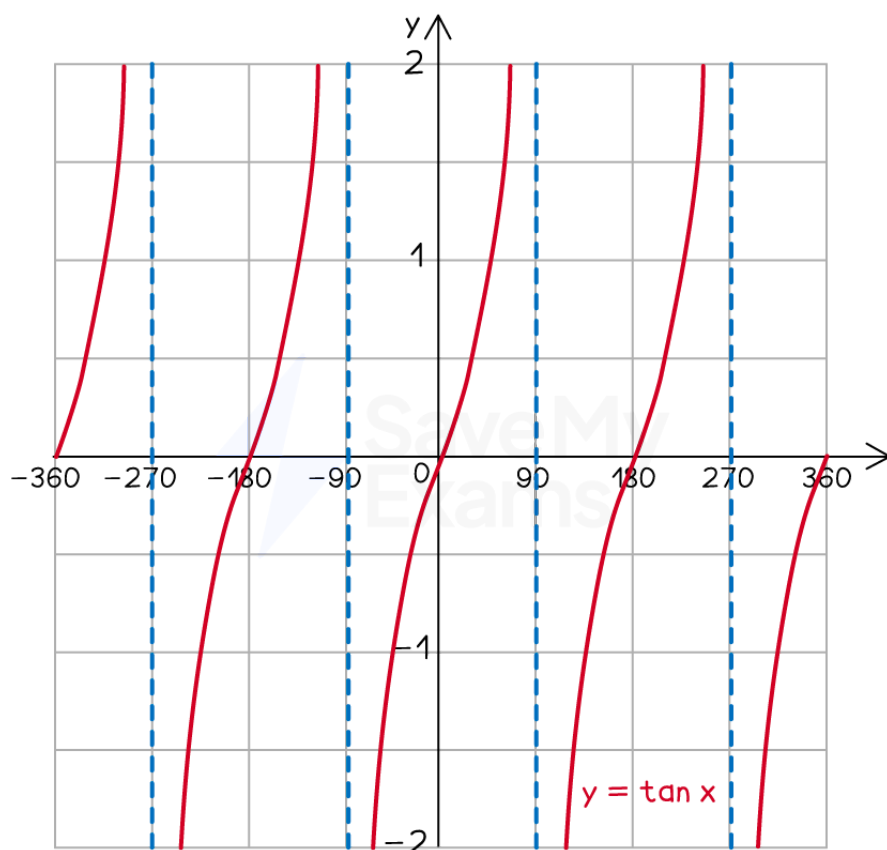
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How do I draw the graph of $y = \tan x$?

- The graph of $y = \tan x$ is **not a wave** but consists of **branches** that **repeat every 180°** (its **period** is 180°)
 - This is **half the period** of $\sin x$ and $\cos x$
- There are **dotted vertical lines** that separate the branches called **asymptotes**
 - These are every 180° at $x = 90^\circ, x = 270^\circ, \dots$
 - The curve cannot touch these, but get **closer and closer** to them
 - A branch starts down at a height of $-\infty$ and goes up to a height of $+\infty$
- $y = \tan x$ goes through the **origin**, $(0, 0)$



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Worked Example

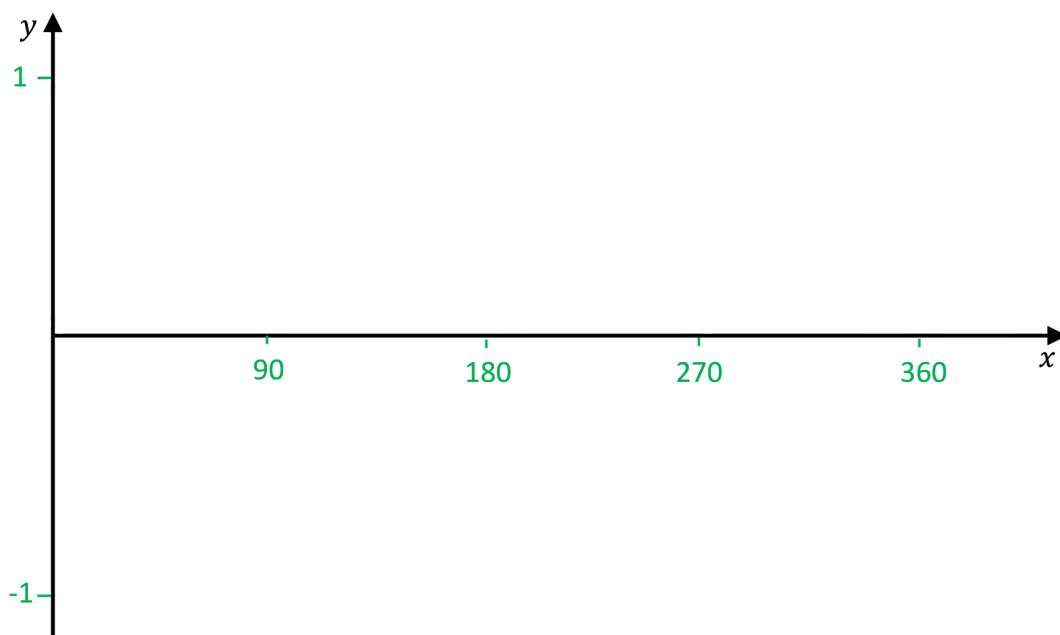
On the axes provided, sketch the graph of $y = \sin x^\circ$ for $0 \leq x \leq 360$.

Mark 1 and -1 on the y-axis

Mark 0, 90, 180, 270 and 360 on the x-axis (try to space them evenly apart)



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$y = \sin x$ starts at $(0, 0)$ then every 90° it cycles through heights of 1, 0, -1, 0, ...

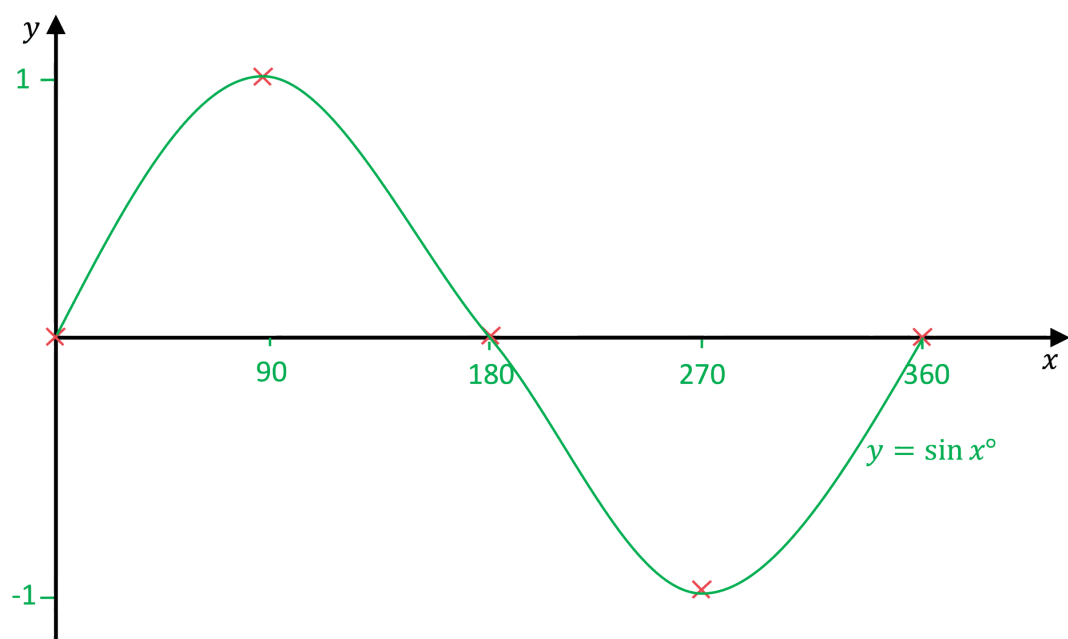
Mark these points on the axes

Join the points with a smooth line

Label the curve with its equation



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Your notes

Solving Trig Equations

Solving Trig Equations

What are trig equations?

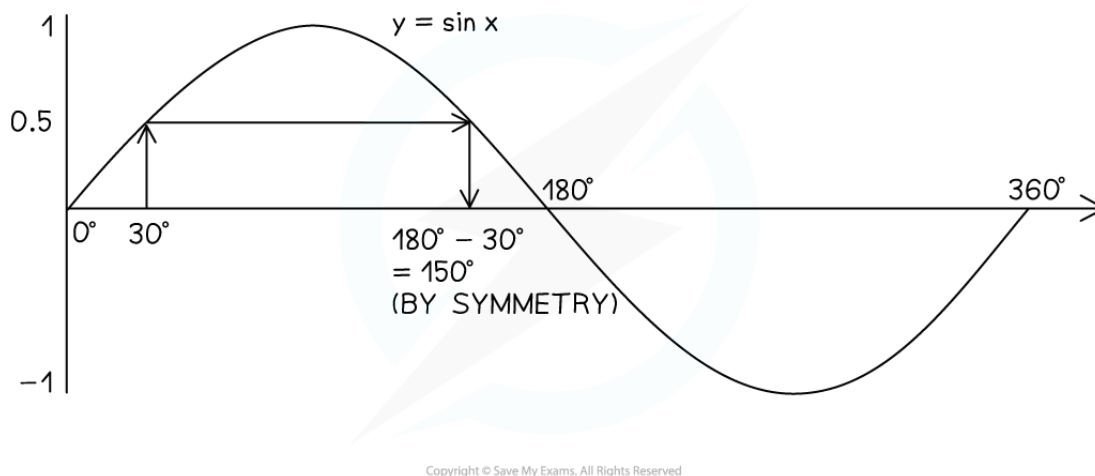
- **Trig equations** are equations involving $\sin x$, $\cos x$ and $\tan x$
- They often have **multiple solutions**
 - A **calculator** gives the **first solution**
 - You need to use **trig graphs** to find the others
 - The solutions must lie in the **interval** (range) of x given in the question, e.g. $0^\circ \leq x \leq 360^\circ$

How do I solve $\sin x = \dots$?

- Find the **first solution** of the equation by taking the **inverse sin function** on your **calculator** (or using an exact trig value)
 - E.g. For the first solution of the equation $\sin x = 0.5$ for $0^\circ \leq x \leq 360^\circ$
 - This gives $x = \sin^{-1}(0.5) = 30^\circ$
- Then **sketch** the **sine graph** for the **given interval**
 - Identify the **first solution** on the graph
 - Use the **symmetry** of the graph to find **additional solutions**
 - E.g. For the equation $\sin x = 0.5$ for $0^\circ \leq x \leq 360^\circ$
 - Sketch the graph $y = \sin x$ for $0^\circ \leq x \leq 360^\circ$
 - Draw on $\sin(30) = 0.5$
 - By the symmetry, the new value of x is $180^\circ - 30^\circ = 150^\circ$
 - The solutions are 30° or 150°



Your notes



- **Check** the solutions
 - E.g. For the equation $\sin x = 0.5$ for $0^\circ \leq x \leq 360^\circ$
 - Substitute $x = 30^\circ$ and $x = 150^\circ$ in to the calculator
 - $\sin(30)$ and $\sin(150)$ both give a value of **0.5**, so are correct
- In general, if x is an **acute** solution to $\sin x = \dots$
 - Then $180 - x$ is an **obtuse** solution to the same equation

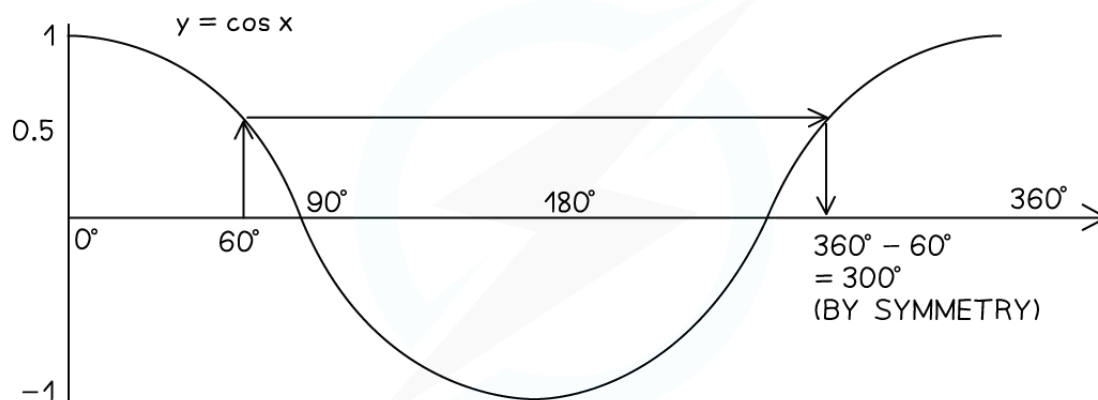
How do I solve $\cos x = \dots$?

- Find the **first solution** of the equation by taking the **inverse cos function** (or using an exact trig value)
 - E.g. For the first solution of the equation $\cos x = 0.5$ for $0^\circ \leq x \leq 360^\circ$
 - This gives $x = \cos^{-1}(0.5) = 60^\circ$
- Then **sketch** the **cosine graph** for the **given interval**
 - Identify the **first solution** on the graph
 - Use the **symmetry** of the graph to find **additional solutions**
- E.g. For the equation $\cos x = 0.5$ for $0^\circ \leq x \leq 360^\circ$
 - Sketch the graph $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$



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- By the symmetry, the new value of x is $360^\circ - 60^\circ = 300^\circ$
- The solutions are 60° or 300°



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- Check** the solutions
 - E.g. For the equation $\cos x = 0.5$ for $0^\circ \leq x \leq 360^\circ$
 - Substitute $x = 60^\circ$ and $x = 300^\circ$ in to the calculator
 - $\cos(60)$ and $\cos(300)$ both give a value of 0.5 so are correct
- In general, if x is a **solution** to $\cos x = \dots$
 - Then $360 - x$ is **another solution** to the same equation

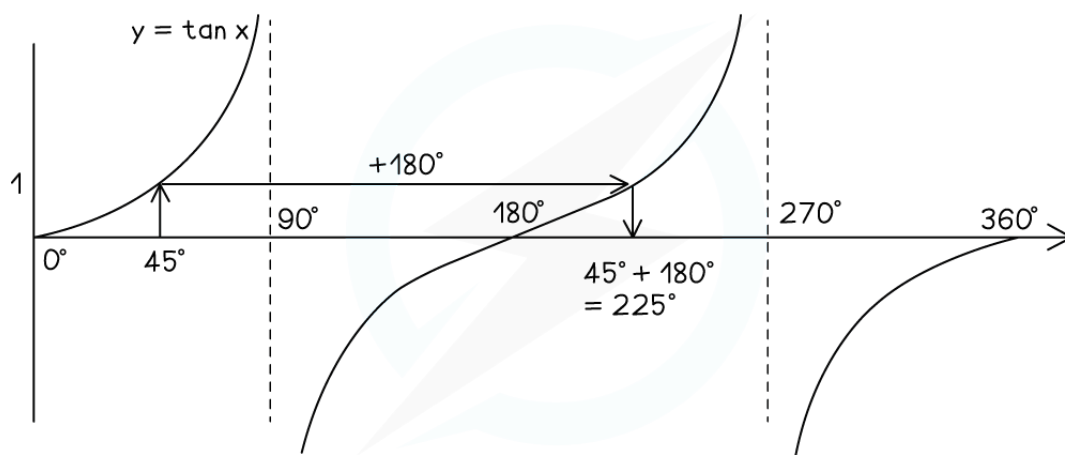
How do I solve $\tan x = \dots$?

- Find the **first solution** of the equation by taking the **inverse tan function** (or using an exact trig value)
 - E.g. For the first solution of the equation $\tan x = 1$ for $0^\circ \leq x \leq 360^\circ$
 - This gives $x = \tan^{-1}(1) = 45^\circ$
- Then **sketch** the **tangent graph** for the **given interval**
 - Identify the **first solution** on the graph
 - Use the **periodic nature** of the graph to find **additional solutions**



Your notes

- E.g. For the equation $\tan x = 1$ for $0^\circ \leq x \leq 360^\circ$
 - Sketch the graph $y = \tan x$ for $0^\circ \leq x \leq 360^\circ$
 - By the periodic nature, the new value of x is $45^\circ + 180^\circ = 225^\circ$



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- **Check** the solutions
 - E.g. For the equation $\tan x = 0.5$ for $0^\circ \leq x \leq 360^\circ$
 - Substitute $x = 45^\circ$ and $x = 225^\circ$ in to the calculator
 - $\tan(45)$ and $\tan(225)$ both give a value of 1 so are correct
- In general, if x is a solution to $\tan x = \dots$
 - Then $x + 180$ is **another solution** to the same equation

How do I rearrange trig equations?

- Trig equations may be given in a **different form**
 - Equations may require **rearranging** first
 - E.g. $2 \sin x - 1 = 0$ can be rearranged to $\sin x = \frac{1}{2}$
- They can then be solved as usual



Your notes

What do I do if the first solution from my calculator is negative?

- Sometimes the first solution given by the calculator for x will be **negative**
 - Continue sketching the graph to the **left** of the x -axis to help
 - Then find **solutions** that lie in the **interval** given in the question



Examiner Tips and Tricks

- Know how to use the **inverse functions** on your calculator
 - It may involve **exact trig values** which do not need a calculator
- **Check** your solutions by substituting them back into the original equation



Worked Example

Use the graph of $y = \sin x$ to solve the equation $\sin x = 0.25$ for $0^\circ \leq x \leq 360^\circ$.

Give your answers correct to 1 decimal place.

Use a calculator to find the first solution

Take the inverse sin of both sides

$$x = \sin^{-1}(0.25) = 14.47751\dots$$

Sketch the graph of $y = \sin x$

Mark on (roughly) where $x = 14.48$ and $y = 0.25$ would be

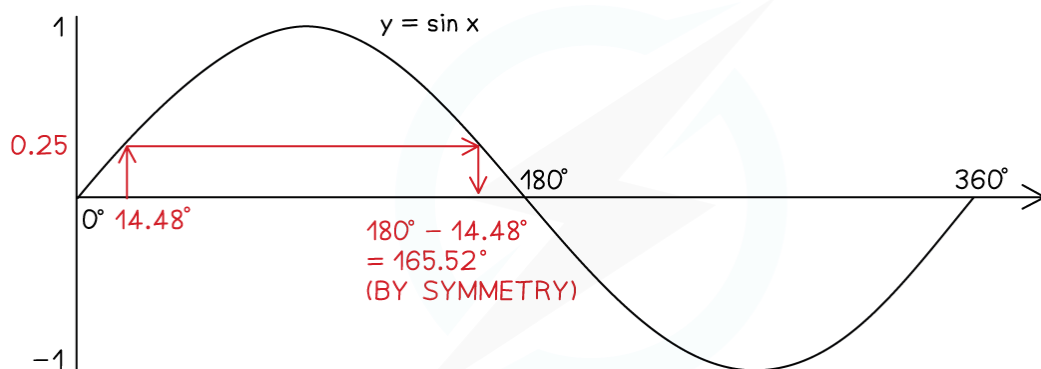
Draw a vertical line up to the curve

Draw another line horizontally across to the next point on the curve

Bring a line vertically back down to the x -axis



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Find this value using the symmetry of the curve

Subtract 14.48 from 180

$$180 - 14.48 = 165.52$$

Give both answers correct to 1 decimal place

$$x = 14.5^\circ \text{ or } x = 165.5^\circ$$